Take-Home Assignment — Multi-View 3D Reconstruction

Nutpaa Technologies Private Limited | Role: Computer Vision Engineer

Part 2 of 2: Take-Home Test | Confidential — Do Not Distribute

Instructions

Provide a fully working and documented implementation for the following tasks. Points awarded based on:

- Tasks completed and correctness of results
- Quality and depth of code documentation
- Ability to explain your code in detail during the live review session
- Profiling output and proposed optimisations
- Clarity of written analysis and extensions

Expected time: 6–8 hours. Partial, well-explained solutions are valued over rushed complete ones. State all assumptions clearly. Submit to Careers@nutpaa.ai — Subject: "Take-Home — CV Engineer — [Your Name]".

Problem Setup

You are given a set of 3D points on a rigid object (a 1m cube, 8 corners at known positions) observed by 3 cameras. All cameras share the same intrinsic matrix K. Camera 1's pose is known (R = I, t = [0, 0, 5]). Camera 2 and Camera 3's poses are unknown. Noisy 2D projections are provided with per-observation confidence scores.

Provided Data (data.json)

Input	Details
8 3D corners	1m cube centred at world origin (metres)
Camera 1	K, R=I, t=[0,0,5] — known. All 8 corners visible. σ=2px noise.
Camera 2	K (same). Pose UNKNOWN. 6/8 visible. σ=2px noise.
Camera 3	K (same). Pose UNKNOWN. 7/8 visible. σ=3px noise.
Confidence scores	Each observation has a 0–1 confidence value.

Tasks

Task 1: Camera Pose Estimation (30%)

Estimate R and t for Camera 2 and Camera 3 using known 3D corners and noisy 2D observations.

- **Implement from scratch:** Minimise OpenCV calls. Each step coded and documented.
- **Occlusion handling:** Skip null observations correctly without interpolation.
- **Documentation:** Explain inner workings of each step inline.

Task 2: 3D Triangulation (25%)

With all 3 poses known, triangulate each of the 8 corners using all views where the point is visible.

- **DLT from scratch:** Construct linear system from projection rows; solve via SVD; explain null-space.
- **Optimal triangulation:** Minimise reprojection error per point (3-variable nonlinear LS).
- **Comparison:** Per-point 3D error (mm) for DLT vs. optimal. Identify highest-error points and explain why.
- **Confidence weighting:** Use data.json confidence scores to weight observations. Evaluate impact.
- **Visualisation:** Plot camera-point geometry. Explain error pattern (near-parallel rays).

Task 3: Robustness Analysis (25%)

Evaluate pipeline robustness under realistic failure conditions.

- **Noise sensitivity:** Repeat pipeline with σ=1,2,5,10 px. Plot pose error and 3D error vs. noise level.
- **Degenerate geometry:** Move Cam3 nearly coplanar with Cam1. Report condition number change and error increase.
- **Outlier rejection:** Inject 2 wrong matches into Cam2. Show vanilla LM fails. Implement RANSAC or Huber/Cauchy loss and show recovery.

Task 4: Documentation & Extension (20%)

Demonstrate engineering maturity and ability to generalise your solution.

- **README:** Setup, run instructions, component explanations, maths, known limitations, assumptions.
- **Profiling output:** Runtime per stage; bottleneck identified; one concrete optimisation proposed.
- **Extension essay (1 page):** "If the object were articulated (e.g. a 5-joint robot arm) instead of a rigid cube, how would you modify this pipeline?" Discuss pose estimation, triangulation, and optimisation changes.

Submission

- Email: Careers@nutpaa.ai
- Subject: "Take-Home — CV Engineer — [Your Name]"
- Include: README.md, profiling output, and extension essay